# TOPS Technologies

## Interview Questions — ST - Introduction and Fundamentals

**Total Questions:** 34

---

**1.** What is software testing and why is it important in software engineering?

---

**2.** Explain the role of testing in the Software Development Life Cycle (SDLC).

> **Hint:** Think about when testing activities start and how they fit between other SDLC phases.

---

**3.** What are some typical testing activities performed during a software project?

> **Hint:** Consider activities from planning to reporting defects.

---

**4.** Draw or describe a simple diagram of software architecture for a basic web application.

> **Hint:** Focus on main components like client, server, and database.

---

**5.** When in a project should testing ideally begin, and why?

> **Hint:** Relate your answer to defect prevention versus defect detection.

---

**6.** List two main objectives of software testing.

---

**7.** Name and briefly describe any two SDLC models.

> **Hint:** Think of models like Waterfall, Agile, Iterative, etc.

---

**8.** What is the difference between the Waterfall and Agile SDLC models?

> **Hint:** Focus on process flow and adaptability to change.

---

**9.** What are entry and exit criteria in software testing?

> **Hint:** These criteria define when testing activities can start or finish.

---

**10.** Give an example of entry and exit criteria for the test execution phase.

---

**11.** How would you map the phases of an SDLC model to a real-world software project?

---

**12.** What is a Software Requirement Specification (SRS) and why is it important for testers?

---

**13.** From a sample SRS document, how would you extract test conditions?

> **Hint:** Focus on identifying what needs to be tested based on requirements.

---

**14.** Write a basic SQL SELECT query to retrieve all records from a table called Employees.

---

**15.** Why should a tester have a basic understanding of OOPS concepts, even if they are not coding?

> **Hint:** Think about how software structure affects testing.

---

**16.** What is the difference between functional and non-functional testing?

> **Hint:** Consider what each type of testing aims to validate.

---

**17.** List and briefly explain any two key principles of software testing.

---

**18.** Given a list of test scenarios, how would you classify them as functional or non-functional tests?

> **Hint:** Functional tests check 'what' the software does; non-functional tests check 'how' it does it.

---

**19.** What are software testing methodologies? Name one and describe when you would use it.

---

**20.** Who is responsible for Quality Assurance (QA) and Quality Control (QC) in a software project, and how do their roles differ?

> **Hint:** QA is process-focused, QC is product-focused.

---

**21.** Explain the difference between testing and debugging.

> **Hint:** Think about who performs each activity and at what stage.

---

**22.** Why is a 'destructive' mindset important for a software tester?

> **Hint:** Consider how this mindset helps find more defects.

---

**23.** Map the main responsibilities of a QA team versus a QC team in a software company.

---

**24.** Describe a scenario where a tester's psychology can impact the effectiveness of testing.

> **Hint:** Think about bias, assumptions, or communication.

---

**25.** What is Agile testing and how does it differ from traditional testing approaches?

> **Hint:** Focus on iterative development and tester involvement.

---

**26.** Explain the purpose of a Scrum Board in Agile testing.

> **Hint:** How does it help teams track progress?

---

**27.** What happens during a Sprint Review in the Scrum process?

> **Hint:** Think about what is demonstrated and who participates.

---

**28.** Create a simple example of a Scrum Board for a testing project with at least three columns.

> **Hint:** Consider columns like 'To Do', 'In Progress', and 'Done'.

---

**29.** When working in Agile, how can testers ensure quality when requirements might change frequently?

> **Hint:** Think about continuous feedback and collaboration.

---

**30.** How would you use an AI tool like ChatGPT to help create test cases from a Software Requirement Specification (SRS), and what would you check before using those test cases?

> **Hint:** Consider both the speed AI offers and the need to verify accuracy.

---

**31.** If you encounter a bug during testing, how could AI-based tools assist you in diagnosing or reporting the issue, and what precautions should you take?

> **Hint:** Think about both automation and the importance of human review.

---

**32.** Integrative: How does understanding the SDLC model in use help a tester define appropriate entry and exit criteria for testing phases?

---

**33.** Integrative: In an Agile project, how would you use both functional and non-functional testing approaches during a sprint?

---

**34.** Integrative: Explain how a tester's role changes between a Waterfall project and an Agile project.

---